Neonatal and Pediatrics Anesthesia

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Part three

POSTANESTHESIA CARE

- PACU recovery handling challenges for very young chldiren
- Hypothermia is most common (infants and young children)
- This is due to inability to regulate body temperature under GA, cold large operating rooms, and continued heat loss
- Minor hypothermia 34-36°C has not been found to affect the recovery period, normothermia should best be restored before discharging a child from PACU

CONT'D

- Severe hypothermia can result in:
  - – Increased MRO_2,
  - – CVS manifestations of hypothermia(arrythmia, high myocardial RO_2)
  - – Prolonged metabolism, and excretion of Anesthetics and
  - – Delayed wound healing

CONT'D

- Emergence agitation refers to:
  - The presence of thrashing,
  - Crying, screaming, and
  - Disorientation accompanied emergence from anesthesia at any age
- This is particularly common in children
- The frequency of this problem varies with age, anesthetics used, and the surgical procedures performed

CONT’D

- More younger children are more at risk like those who had strabismus or adenoidectomy surgeries
- There is a higher incidence of agitation with the use of pure sevoflurane anesthesia
- This phenomenon is not strictly related to pain

CONT'D

- It happens with effective treatment with a variety of medications including fentanyl, dexmedetomidine, midazolam, and ketamine
- Low incidence is clearly with propofol-based anesthesia
- The behaviors stop within 30 to 45 minutes after surgery

CONT'D

- Nausea and vomiting occurs frequently after eye or ear surgery
- Studies found, the problem is even more widespread (39 to 73%)
- The physiological mechanisms for postoperative nausea:
  - Central causes (opiates)
  - Gastrointestinal malfunction (ileus or gastric distention),
  - Surgical procedure itself and
  - postoperative pain

CONT’D

- Controlling this problem is not only for PACU personel, It begins with selection of agents or techniques used for anesthesia
- E.g, using induction and maintenance of anesthesia with propofol-air-oxygen (about 23% incidence of N/V)
- Than using halothane-N20-droperidol (about N/V 50% incidence) for strabismus surgery

CONT'D

- Pretreatment with ondansetron-0.15 mg/kg, droperidol-0.075 mg/kg, or metoclopramide-0.15 mg/kg, has been very successful in reducing N/V for patients at higher risk
- Prophylactic use of ondansetron(SSRB), is particularly effective in reducing postoperative nausea and vomiting
- But it is more expensive than other antiemetics

CONT'D

- Pain mgt in the PACU should get special attention
- Physiological responses like tachycardia, hypertension, N/V, and agitation are indicators of analgesic need in younger children than pain self-report(best pain ass't modality) in older children
- IV opiates like fentanyl or morphine are used most commonly to treat moderate to severe pain in PACU
- This drugs should be carefully titrated(untoward effects)

CONT'D

- Children may be terrified in PACU for strange place with new people and may be disoriented from residual effects of anesthesia
- Some children may experience nightmares, or have behavioral problems after a surgery
- Measures taken to calm and comfort the child may reduce the incidence of these sequelae and aid in the overall recovery

CONT'D

- Many institutions have found that allowing parents to soothe the child during recovery is beneficial

Recommendations for patient assessment and monitoring in PACU

- **Respiratory**
- Assessment of airway patency, respiratory rate, and oxygen saturation should be periodically performed
- Give due emphasis to monitoring oxygenation and ventilation

CONT’D

- **Cardiovascular**
- Heart rate and blood pressure should be routinely monitored
- Electrocardiographic monitors should be immediately available

Neuromuscular

- Assessment of neuromuscular function should be performed for all patients who received NDMRs or who have medical conditions associated with neuromuscular dysfunction
- **Mental Status**
- Mental status should be periodically assessed.
- **Temperature-** Patient temperature should be periodically assessed (mainly for very young)

CONT'D

- **Pain-** should be periodically assessed and treated
- **Nausea and Vomiting**
- Periodic assessment of postoperative nausea and vomiting should be routinely performed

CONT'D

- **Hydration**
- Postoperative hydration should be assessed and managed
- Certain procedures may involve significant blood loss and require additional intravenous fluids management.

CONT'D

- **Urine**
- UOP assessment and of urinary voiding should be performed on a case-by-case basis for selected patients or selected procedures
- **Drainage and Bleeding**
- Assessment of drainage and bleeding should be performed periodically as need

Post-operative Upper Airway Obstruction

- **LOSS OF PHARYNGEAL MUSCLE TONE**
- Usually in **sedated or obtunded** patients
- The persistent effects of inhaled and intravenous anesthetics, neuromuscular blocking drugs, and opioids all contribute to the loss of pharyngeal tone in the PACU patient

CONT'D

- **RESIDUAL NEUROMUSCULAR BLOCKADE**
- Expect residual NMB at any time in PACU as the diaphragm recovers from neuromuscular blockade before the pharyngeal muscles do

CONT'D

- **LARYNGOSPASM**
- It typically occurs in the transitional period when the extubated patient is emerging from GA yet not fully awake
- Patients who are asleep during PACU handling are also at risk
- It is due to AW irritants like secresions, blood, AW edema
- Remedy- Remove the secretions and apply Jaw trust and CPAP of 5-20 cm H2O

CONT'D

- If not relieved- add 0.3-0.5mg/kg propofol IV, then 0.1-0.8mg/kg of sux IV or 3mg/kg Sux Im
- Then use NPA, OPA or LMA to aid in ventilation if that relieves
- Look for the cause if opiod or bnzos...look for reversal
- If still not relieved- use full dose anesthesia and muscle relation to allow for tracheal intubation

Pulmonary Edema

- Pulmonary edema in the immediate postoperative period is often cardiogenic in nature, secondary to intra vascular volume overload or congestive heart failure
- The other causes are postobstructive pulmonary edema (secondary to airway obstruction), sepsis, or transfusion (transfusion-related acute lung injury-TRALI), may occur less frequently

CONT’D

- **Monitoring and Treatment of Hypoxemia**
- With continous standard pulse oxymetry monitoring in the PACU, administer supplemental oxygen only for those need it
- Immediate desaturation in a PACU correlated positively with patient age, body weight, ASA status, GA, high IV fluid >1500 mL, duration of anesthesia and female gender

CONT’D

- **Hemodynamic Instability**
- A change in vitals of patients in a PACU like systemic HTN, hypotension, tachycardia or bradycardia alone or in combination
- This change has a negative impact on long-term outcome
- POP systemic HTN and tachycardia are associated with increased risk of unplanned ICU admission and a higher mortality than hypotension and bradycardia

CONT'D

- **Cardiac Arrhythmias**
- POP arrythmias are frequently transient and multifactorial
- The reversible causes include hypoxemia, hypoventilation and associated hypercapnia, pain, agitation, anesthetic agents, medications (β blockers, opioids)endogenous or exogenous catecholamines, electrolyte abnormalities, acidemia, fluid overload, anemia, and substance withdrawal

CONT'D

- **Renal Dysfunction**
- Prerenal, intrarenal, and postrenal etiologies
- POP the frequent causes include intraoperative insult exacerbating a preexisting renal insufficiency
- In PACU, diagnostic efforts should focus to identify and treat the reversible causes of oliguria (UOP<0.5 mL/kg/h)

CONT'D

- Confer with the surgical team regarding the details of the surgical procedure (urologic, gynecologic) to r/o anatomic obstruction or disruption of the ureters, bladder, or urethra
- Patient-related factors like preexisting renal insufficiency (CKD), DM, HTN, morbid obesity, history of steroid use, male sex, and old age, are risks considered for peri-operative renal failure

CONT’D

- **Postoperative Hypothermia and Shivering**
- POP hypothermia, defined as a core temperature <36°C
- Detrimental /unpleasant condition occuring after GA or RA
- ASA guidline states that patient’s temperature should be measured within 15min after end of anesthesia and ideally be at least 36°C

CONT'D

- Postoperative shivering often occurs after GA and RA
- The incidence of postoperative shivering may be as high as 66% after GA
- Identified risk factors include young age, endoprosthetic surgery, and core hypothermia

CONT'D

- Mechanism of hypotherimia are heat loss from radiation, convection, evaporation, and conduction
- Shivering at POP is usually from hypothermia or thermoregulatory mechanism
- Non-simultaneous recovery of spnial cord and brain, spinal cord fast recovery from GA-clonic activities (doxapram)

CONT'D

- Other causes are action of kappa opioid, NMDA, and 5-hydroxytryptamine receptors
- High incidence of postanesthetic shivering in patients who receive high-dose remifentanil is due to hyperalgesia in these patients from sudden opioid withdrawal resulting in the stimulation of NMDA receptors

CONT’D

- A small dose of intraoperative ketamine reduced the incidence of remifentanil-induced POP shivering
- Tramadol, a weak μ-opioid receptor agonist, and NA and serotonin reuptake inhibitor, is effective in preventing postoperative shivering while also contributing to analgesia

CONT'D

- **Treatment**
- A number of opioids, ondansetron, clonidine, and ketamine are shown to be effective to treat shivering
- Meperidine, 12.5-25 mg IV, is most commonly used in adults
- Intraoperative infusion of dexmedetomidine has been shown to be an effective prophylactic measure
- **What are the clinical implications if left untreated?**

CONT'D

- **Delayed Awakening**
- Defined as non-responsive to stimulations in 60-90min even after prolonged surgery and anesthesia
- If no emergence takes place at that point, consider:
- Residual drug effects after too much anesthetics (most frequent cause)

CONT'D

- Susseptability to side effects of certain medications due to age, underlying disease, or metabolic derangements
- BDZs, opioids, and NMBDs are the most common drugs
- However, after a very long anesthetic, propofol and volatile anesthetics can also cause a delay in emergence
- The central anticholinergic syndrome (CAS) my cause a delay

CONT'D

- Hypothermia (<33°C), hyponatremia, hypercalcemia, hypermagnesemia), hypo- or hyperglycemia
- Underlying metabolic diseases (e.g., liver, kidney, or thyroid abnormalities) can delay emergence
- Cerebral hypoxia, seizures, elevated ICP, or any intracerebral event (hemorrhage, thrombosis, embolus) should be considered

CONT'D

- In any patient with a delayed emergence, AW, breathing, and circulation should be assessed
- Confirm that all anesthetic agents are D/C
- Checked patient's body temperature (pedi) upon arrival in PACU
- If there is hypothermia, active rewarming should be done
- A cardiopulmonary and neurologic exam (including pupils, cough and gag reflex, motor/strength) should be done

CONT’D

- If there is residual NMBDs neostigmine/glycopyrrolate or sugammadex is used
- If a residual opioid effect is suspected
- Naloxone in small increments (40 μg every 2 minutes up to 200 μg) can be titrated to effect
- If residual BDZs effect is suspected, flumazenil in 0.1 to 0.2 mg increments every 1 minute up to 1 mg can be titrated to effect

CONT'D

- Check blood glucose level and treat it by dextros if hypoglycemia(most common in pediatrics) and by insuline if hyperglycemia is suspected
- An ABG and electrolyte panel should be obtained and
- Treat a CO_2 narcosis by hyperventilation (potentially ETT)
- Any electrolyte disturbances should be corrected
- If none of the above strategies work, consider CAS and

CONT'D

- Simultaneosly consulting a neurologist is important to rule out any cerebrovascular accident obtaining a stat head CT
- If the patient still does not emerge, admission to the ICU for further monitoring and serial exams should be initiated

CONT'D

- **Discharge Criteria**
- Specific PACU discharge criteria may vary
- General principles are universally applicable (Activity, breathing, circulation, consiousness and color /saturation)
- A mandatory minimum stay in the PACU is not required
- Patients must be observed until they are less risky for ventilatory depression and their mental status has returned to a baseline

CONT'D

- Hemodynamic criteria are based on the patient's baseline parametrs without specific systemic BP and HR requirements
- An assessment and written documentation of the patient's peripheral nerve function on discharge from the PACU
- In the years after Aldrete abd kroulik deveoloped a score for discharge criteria in 1970, the advent of technology and ambulatory surgery made that to be modifiied (next slide)

Modified discharge criteria from a PACU

Variable evaluated	score
ACTIVITY	
Able to move four extremities on command	2
Able to move two extremities on command	1
Able to move no extremities on command	0
BREATHING	
Able to breathe deeply and cough freely	2
Dyspnea	1
Apnea	0
CIRCULATION	
Systemic blood pressure ≤20% of the preanesthetic level	2
Systemic blood pressure is 20% to 50% of the preanesthetic level	1
Systemic blood pressure ≥50% of the preanesthetic level	0
CONSCIOUSNESS	
Fully awake	2
Arousable	1
Not responding	0
OXYGEN SATURATION (PULSE OXIMETRY)	
Greater than 92% while breathing room air	2
Needs supplemental oxygen to maintain saturation >90%	1
Less than 90% with supplemental oxygen	0

CARDIOPULMONARY RESUSCITATION

Introduction

- Brain ATP is depleted after 4-6 min of no blood flow
- It replanishes within 6min of starting effective CPR
- ABCDEs electrical are the major components of resuscitation
- Two mechanism of blood flow during closed-chest compression have been suggested, cardiac pump and thoracic pump

CONT'D

- Myocardial perfusion is 20 to 50% of normal
- Cerebral perfusion is maintained at 50 to 90% of normal
- CPR has limited success, with only about 40% of victims being admitted to the hospital and 10% surviving to discharge

CONT'D

- Adequacy of closed-chest compression is usually judged by palpation of a pulse in the carotid or femoral vessels
- ETCo2 is an excellent noninvasive guide to the adequacy of closed-chest compressions
- Effective uninterrupted chest compressions and defibrillation, if any, should be done before medications

CONT'D

- After vasopressors, drugs most likely to be of benefit during CPR are those that help suppress ectopic ventricular rhythms
- Ventricular fibrillation is the most common ECG pattern found during witnessed sudden cardiac arrest in adults

CONT'D

- Untreated ventricular fibrillation is a time-sensitive model with three phases: electrical, circulatory, and metabolic
- Arrest is less likely to be a sudden event and more likely related to progressive deterioration of respiratory and circulatory function in the pediatric age group
- Tx of cardiac and respiratory arrest is integral part of anesthesia practice

CONT'D

- Airway and ventilation problems lead to asystole and PEA as the most common presenting rhythms
- Sequele of myocardial and cerebral ischemia are like for the adult and the basic approach to the unresponsive victim is similar
- Specific anatomic and physiologic considerations (pedi)
- Though defibrillation is less frequently needed in children, starting energy is 2 J/kgrecommended , can be doubled if it is unsuccessful

CONT'D

- The main focus of this session is on a systematic [Airway (A), Breathing (B), Circulation (C), Disability (D), Environment - Exposure (E)] approach to the immediate assessment and treatment of critically ill children and children in cardiorespiratory arrest

CONT'D

- The aims and principles of this ABCDE approach are:
  - To identify potentially life-threatening conditions and to deliver life-saving Tx in a simultaneous and continuous manner in order to prevent further deterioration towards cardiorespiratory arrest
  - To standardise initial assessment and management with a priority-based approach in order to enhance reliability and improve team performance

Children difer

- Size and weight (kg)=(age(yr)+4)*2
- If estimated weight is above 50 kg, consider to use adult dosing schemes for calculation or use 50 as upper limit

Anatomical and physiological diferences

Airway (A)

- Large head is usually compared to the body(supine AWO)
- Small face and mouth and relatively large tongue easily obstructs the airway in the unconscious child
- Easily compressible foor of the mouth and UAWO by pressure on soft tissues under the chin

CONT'D

- Preferential nasal breaathers upto 5-6 months
- Nasal obstruction in newborns/infants (e.g. mucus secretions due to respiratory infections, anatomical abnormalities, NG tubes) can increase the work of breathing and lead to respiratory failure
- ATH is a relatively common finding b/n age of 2-6yrs
- It may contribute to UAWO(difculties with bag-mask ventilation)

CONT'D

- Larynx is located at C2 in the infant compared with C4-C5 for older children and adults
- U-shaped epiglottis, protruding towards the pharynx at an angle of 45°; the vocal cords are short
- In children < 8 years, funnel shaped larynx with narrowest section at the level of the subglottic cricoid cartilage

CONT'D

- In older children, it is cylindrical
- Especially in infants the narrow airway makes it particularly vulnerable to obstruction by e.g. oedema

Breathing (B)

- Immature lungs at birth with an air-alveolar interface of $3m^2$ compared with the adult of $70\ m^2$
- About tenfold increase in number of small airways from birth to adulthood
- At the birth of premature newborns, alveolar opening may be impeded by surfactant defciency; which may require surfactant administration to assist with alveolar opening and patency

CONT'D

- The mechanics of breathing alter with age
- The diaphragm is the main respiratory muscle in children
- Mechanical obstacles to diaphragmatic contraction of:
  - Abdominal origin (gastric distension, pneumoperitoneum, intestinal occlusion) or
  - Pulmonary origin (hyperinfation, e.g., with bronchiolitis, asthma or FB inhalation) can lead to inefective ventilation

CONT’D

- More easily collapsing smaller lower airways in infants with increased intrathoracic pressure due to forced expiratory efort
- Soft and pliable ribs and the intercostal muscles are relatively weak and inefective compared to the diaphragm in infants

CONT'D

- In older children, those muscles are more developed and contribute well to the mechanics of ventilation
- The ribs are ossifed and supports as well as forms a rigid structure with less likely to collapse in respiratory distress
  - The intercostal and sternal recession in older children is indicative of more serious respiratory failure than it is in infants

CONT'D

- Due to high metabolic rate and oxygen consumption,RR is higher in infants and young children than in adults
- Work of breathing can account for up to 40% of CO in children
- Children have a smaller FRC and thus less respiratory reserve
- Desaturation will occur much faster
- This is even more obvious when in supine position

Circulation (C)

- The circulating volume of the neonate is 80 ml/kg, decreasing to 60-70 ml/kg in adulthood
- Children are more susceptible to fuid loss or relative dilution, as the total volume is low
- Due to high metabolic rate and CO, HR is higher in children than in adults

CONT'D

- The SV cannot increase in the same way as in adults, CO in children is mainly determined by HR
- BP in turn is lower in infants and small children than in larger children or adolescents
- Normal Values for Respiratory Rate, Heart Rate and Blood Pressure in infants and children (see next slide)

Age	Normal RR (upper limit)	Normal HR (upper limit)	Normal BP syst. (lower limit)	Normal BP mean (lower limit)
1 month	35 (55)	120 (175)	60 (50)	45 (35)
1 year	30 (40)	110 (170)	80 (70)	55 (40)
2 year	25 (30)	100 (160)	90 + 2 *age (70 + 2 *age)	55 + 1.5 *age (40 + 1.5 *age)
6 year	20 (25)	90 (130)	90 + 2 *age (70 + 2 *age)	55 + 1.5 *age (40 + 1.5 *age)
12 year	15 (20)	80 (100)	120 (90)	80 (65)

Pathways leading to cardiorespiratory arrest

- Cardiorespiratory arrest in children is very rare than in adults
- The causes of cardiorespiratory arrest are also not the same in children and in adults
- These diferences change with age, from the newborn and the neon9atal period through infancy and childhood (including adolescence) until adulthood.

CONT’D

- Cardiac arrest in children is less likely due to primary cardiac disease (primary cardiorespiratory arrest)
- This one is more frequent in the adult population and has an acute and unpredictable onset refecting intrinsic heart disease
- Immediate defibrillation is usually required as often the terminal rhythm will be a shockable one (VF or pVT)
- In these situations, every minute delay in defibrillation will signifcantly decrease the probability of ROSC

CONT'D

- Secondary cardiorespiratory arrest is much more common in children, refecting the inability to deal with certain triggering injuries/illnesses
- Here cardiac arrest is consequence of severe tissue hypoxia causing myocardial dysfunction
- Tissue hypoxia can arise from respiratory failure or from severe hypoperfusion as in circulatory failure

CONT'D

- Initially adaptive physiological responses are activated to protect the heart and the brain from hypoxia and compensate for the respiratory or circulatory failure
- With the progression of failure or injury, however, the body fails to sustain these responses

CONT'D

- This represents a decompensated physiological state, either respiratory or circulatory depending on the underlying cause; both can also be present simultaneously
- Respiratory and circulatory failure combine as the child's condition worsens, leading to cardiorespiratory failure and then cardiorespiratory arrest

CONT'D

- The terminal rhythm will often be a non-shockable one: pulseless electrical activity [PEA] or asystole
- The outcome of resuscitation from this cardiorespiratory arrest is often poor,particularly if the duration of the underlying cardiorespiratory failure and the subsequent arrest is prolonged

CONT’D

- Early recognition of signs of failure and carrying out efective interventions are essential
- Identifying and resuscitating a child with respiratory arrest, but whose heart is still beating, is associated with at least 50 % long term survival and good neurological function,
- But the survival without neurological sequelae for a child after an asystolic arrest is much lower, even for high income countries

CONT'D

- Hence,we emphasize on early recognition of at risk situations and prioritise the frst steps of an efective management with the aim of preventing further deterioration

Recognition of the critically ill child

Introduction

- 11 million die each year before reaching age of 5-year
- 99% are from low and middle income countries (LMIC)
- 75% of deaths are from preventable or treatable causes such as pneumonia, diarrhoea, malaria, and measles
- They can become unwell very quickly, and the outcome from cardiac arrest in a child is poor
- Hence, early recognition and treatment of ill child is vital

CONT'D

- In developed nations recognition, assessment and management of critically ill child has improved after
- Courses like APLS in the UK and
- Paediatric Advanced Life Support (PALS) in the US
- These courses are now often mandatory for clinicians working with children

CONT'D

- WHO has developed the Emergency Triage Assessment and Treatment (ETAT) system to reduce childhood mortality, mainly within the first 24 hrs of admission
- About 50% of children who die after hospital admission do so in the first 24 hours
- ETAT is developed to include admission care for the first 24 hours

PRINCIPLES OF MGMT OF CRITICALLY ILL CHILD

- Knowing normal physiological values across ages, signs of illness, and how a child compensate for critical illness is important
- Understanding the principles of triage and the ‘ABC’ approach to assessment and treatment is important
- The normal values of heart rate, respiratory rate and systolic blood pressure are discussed

Physiological compensation

- Children can compensate effectively during the early stages of serious illness (masks how well they really are)
- Compensation is the ability to maintain perfusion of ‘vital’ organs such as the brain and heart at others expense
- Decompensation is failure of this compensation
- If not addressed it rapidly leads to organ failure and death

CONT'D

- Signs of compensation are high RR, HR and peripheral vasoconstriction causing cold extremities
- Health workers must be vigilant to signs of compensation so that intervention and treatment can be started promptly
- Signs of decompensation include hypotension and bradycardia, and babies may develop apnoeic episodes

CONT'D

- Assessment of end-organ function is important and can also indicate decompensation
- E.g, children may appear outwardly well, but they are listless, not interested in their surroundings and tolerate examination and interventions without complaint; this can be an early sign of neurological decline or fatigue

CONT'D

- Confusion -a worrying sign and shows low cerebral perfusio
- AVPU is used to assess level of consciousness
- Low urine output due to inadequate renal perfusion is often a late sign in children (unlike adults)
- Hence, not useful in initial emergency care
- But, if a mother reported that a child was unuric, this is a serious

PREPARATION

- When preparing to receive a sick child to your facility, individual roles and responsibilities of the resuscitation team members should be clearly understood
- The **WETFLAG** mnemonic (Weight, Energy, Tube, Fluids, Adrenaline, Glucose) is helpful to prepare in receiving a sick child to the hospital facility (see below)

WETFLAG' mnemonic

Weight (this estimate is guide only)

- 0-12 months = (0.5 x age (months)) + 4
- 1-5 years = (2 x age) + 8
- 6-12 years = (3 x age) + 7

Energy = 4 x weight (J)

- (Energy required for defibrillation)

Tube = age/4 + 4

- (Approx. size of uncuffed tracheal tube)

CONT'D

Fluids = 20ml.kg-1 of 0.9% saline

- (Fluid bolus for shocked child)

Adrenaline = 10mcg/kg= 0.1ml/kg of 1:10,000

- (Dose in cardiac arrest)

Glucose=2ml/kg-10% glucose

- (Treatment of hypoglycaemia)

CONT’D

- A worked example of the WETFLAG calculations for a 5-year-old child and < 1year is given below

WETFLAG calculations for a 5-year-old child

- **Weight** = (2 x 5) +8 = 18kg
- **Energy** = 4 x 18 = 72J
- **Tube** = 5/4 + 4 = 5 - 5.5 - 6
- **Fluids** = 20ml x 18 = 360ml of 0.9% saline
- **Adrenaline** 0.1ml.kg-1 = 1.8ml 1:10,000
- **Glucose** = 2ml x18 = 36ml 10% glucose

WETFLAG cal. for a child <1 yeal, Lorazepam

Infant	Weight (KG)	Energy	ET Tube (mm)	Fluid	Lorazepam	Adrenaline (ml)	Glucose (ml)
Months	(0.5 x Age) + 4	4j/KG	Uncuffed (Cuffed)	20ml/kg NaCL 0.9%	0.1mg/kg	0.1ml/kg 1:10'000	2ml/kg 10%
0	4	16	3-3.5 (3)	80	0.4	0.4	8
1	4.5	18	3-3.5 (3)	90	0.45	0.45	9
2	5	20	3-3.5 (3)	100	0.5	0.5	10
3	5.5	22	3-3.5 (3)	110	0.55	0.55	11
4	6	24	3-3.5 (3)	120	0.6	0.6	12
5	6.5	26	3-3.5 (3)	130	0.65	0.65	13
6	7	28	3-3.5 (3)	140	0.7	0.7	14
7	7.5	30	3-3.5 (3)	150	0.75	0.75	15
8	8	32	3-3.5 (3)	160	0.8	0.8	16
9	8.5	34	3-3.5 (3)	170	0.85	0.85	17
10	9	36	3-3.5 (3)	180	0.9	0.9	18
11	9.5	38	3-3.5 (3)	190	0.95	0.95	19
12	10	40	3-3.5 (3)	200	1	1	20

Note

- Fluid for trauma= 10mlkg of 0.9%Nacl x 4. then transfusion
- Always escalate up defi energy joules (150, adult biphasic)
- Drugs are maximum drug calculations-use clinical judgement
- never forget the FGH- Fluids, glucose and hypotermia

TRIAGE AND THE ABCD+ EFG APPROACH

- Triage is system of prioritising who needs treatment first
- Initial triage involves categorizing children who present to the emergency department into three groups of urgency:
- Emergency cases,
- Priority cases and
- Non-urgent cases

Traige categories for five year old child

- **Emergency cases** →Immediate emergency treatment- ABCD
- **Priority cases** → Assessment and rapid attention-3TPR-MOB’
- **Non-urgent cases** → Can wait their turn in the queue

Recognition of the critically ill child

- Like any other medical emergency, a systematic and priority-based approach enhances reliability and improves communication and optimisation of teamwork

CONT’D

- For all children, emergency ass’t includes 4 sequential steps:
    1. First Observational Assessment or Quick Look
    2. Primary Physiological Assessment by using the ABCDE+ FGH approach
    3. Secondary Clinical Assessment with a focused medical history and detailed physical examination
    4. Tertiary Complementary Assessment with laboratory, imaging and other ancillary studies

CONT’D

- The most crucial element of this systematic approach is the principle that when a potentially life threatening problem is identifed immediate treatment is performed before moving on to the next step in the sequence: ‘Treat as you go’ approach

A quick look

- The components of the Quick Look ass’t are the ‘BBB’: [Behaviour – Breathing – Body colour]
- Each component is evaluated separately
- Any abnormality noted denotes an unstable child who needs some immediate clinical intervention and a hands-on ABCDE primary assessment

Behaviour

- Abnormal signs of behaviour include:
- No spontaneous movement, unable to sit or stand
- Less alert or engaged with clinician or caregiver; not resisting examination
- No interaction or eye contact with people and environment, toys and objects

CONT’D

- Being inconsolable, unable to comfort
- Has a weak cry; abnormal speech for age
- Abnormal positioning; preference for seated position
- Having convulsions; abnormal movements

BREATHING

- Describes the child's respiratory status (oxygenation and ventilation status)
- **Abnormal breathing sounds:** snoring, mufled or hoarse speech, stridor, grunting, wheezing
- **Retractions:** supraclavicular, intercostal; head bobbing
- **Flaring** of the nares on inspiration

Body colour

- Describes the child’s circulatory function, primarily in terms of skin perfusion
- **Pallor:** white or pale skin or mucous membranes
- **Mottling:** patchy skin discoloration due to various degrees of vasoconstriction
- **Cyanosis:** bluish discoloration of skin and mucous membranes

CONT'D

- **The Apgar score**, recorded at 1 min and again at 5 min after delivery, is the most valuable ass't of the neonatal rescucitation
- The 1-min score correlates with survival
- The 5-min score is related to neurological outcome (next slide)
- Neonates with Apgar scores of 8–10 are vigorous and may require only gentle stimulation (flicking the foot, rubbing the back, and additional drying)

	Points		
Clinical sign	0	1	2
Heart rate (beats/min)	Absent	< 100	> 100
Respiratory effort	Absent	Slow, irregular	Good, crying
Muscle tone	Flaccid	Some flexion	Active motion
Reflex irritability	No response	Grimace	Crying
Color	Blue or pale	Body pink, extremities blue	All pink

Primary ass’t-ABCDE

- It is important to repeat periodically the primary ass’t
- When a child was unresponsive at frst impression, a brief check of responsiveness should be performed before starting with the airway assessment by gentle verbal or tactile stimulation to establish a more exact level of consciousness

Airway (A)

- AW is assessed to see its patency
- This is done by "look, listen, feel":
- **LOOK** for chest wall (& abdominal) movements
- **LISTEN** for breath sounds and noises at the mouth and nose (or chest auscultation)
- **FEEL** for air fow at the mouth and nose

CONT'D

- If the airway is not patent or at risk, immediate interventions are necessary before proceeding to the breathing assessment
- These interventions are suctioning, airway opening manoeuvers /adjuncts

Breathing (B)

- Failure to maintain ventilation is mainly due to low RR (e.g., in an opioid overdose) or a low VT (e.g. airway obstruction, neuromuscular disease)
- Failure of gas exchange at the alveolar-capillary membrane is usually a result of fuid accumulation within the alveoli, and results in a fall of arterial oxygen levels and increased lung stifness (decreased lung compliance)

CONT’D

- Reduced O2 levels stimulate the respiratory centre and the RR increases; at the same time, the work of breathing increases because of the increased rate and the increased stifness
- Breathing assessment can be done by evaluating:
- **R**espiratory rate, **W**ork of breathing , **T**idal volume, **O**xygenation

CONT'D

- By doing so, the respiratory status can be categorized as stable or as compensated or decompensated respiratory failure
- This categorisation should always be based on a constellation of fndings and not on a single abnormal element
- If there is any it should be managed before moving to the circulation ass't

Respiratory Rate

- Abnormalities of the RR can be classifed as:
  - Too rapid (tachypnoea)
  - Too slow (bradypnoea), or
  - Absent (apnoea)

Work of breathing

- High work of breathing in children is manifested as: intercostal, sternal, and subcostal retraction (or recession), nasal faring, head bobbing and contraction of the anterior chest wall muscles
- In children > 5 years, their chest is less compliant, retractions indicate severe impairment of the child’s respiratory function
- Head bobbing (use of sternocladomastoid muscle) & see-saw (paradoxical mov’t abdomen and chest wall) respiration

Tidal Volume (VT)

- Spontaneous tidal volume stays constant throughout life at around 7 ml/kg.
- It can be qualitatively assessed by:
  - Visual detection of chest expansion adequacy
  - Listening to air entry in all zones of the lungs (auscultation)
- A ‘silent’ chest is an ominous sign that indicates a dramatically reduced tidal volume

Oxygenation

- A more reliable way to determine oxygenation is by measuring the oxygen saturation with a pulse oximeter when respiratory failure is suspected, even in the absence of cyanosis
- A reading of 95% on air is acceptable,but it is a serious finding in a child on 60% oxygen
- Oximeters are less accurate when SpO2 is < 70%, in low peripheral perfusion states (e.g. shock, hypothermia) and/or in the presence of carboxyhaemoglobin or methaemoglobin

CONT'D

- **Efects of respiratory failure on other organs**
- Tachycardia
- A changing LOC is indicative of decompensation
- Agitation (like fighting the oxygen mask) or drowsy
- The final event will be loss of consciousness

Circulation(C)

- **Circulatory Failure**
- **Shock** is a clinical state in which blood fow and the delivery of tissue nutrients do not meet tissue metabolic demand
- **Compensated shock** -early phase of shock before HPTN occurs
- Signs of compensation are tachycardia, poor peripheral perfusion (cool extremities, prolonged capillary refll time), weak peripheral pulses and decreasing urinary output

CONT'D

- **Decompensated shock** -is present when hypotension develops and vital organ (heart, brain) perfusion is compromised as manifested by decreased LOC and weak central pulses
- The aim in the management of shock is to prevent the onset of decompensated shock, as this may lead to **irreversible shock and death**

Cardiovascular relationships

- **Preload**
- Contractility
- Afterload
- **Stroke Volume**/Heart Rate>CO
- CO/SVR>>Blood pressure

CONT'D

- Circulation assessment is done by evaluating the 5Ps:
  1. **P**ulse-Heart Rate
  2. Peripheral **P**erfusion
  3. **P**ulses volume
  4. Blood **P**ressure
  5. **P**reload
- Circulatory status can be categorized as normal, compensated or decompensated circulatory failure (shock)

Pulse-Heart Rate [HR]

- Sinus tachycardia is a common response to anxiety, fever or pain but is also seen in hypoxia, hypercapnia or hypovolemia (non-specifc but early sign)
- If his cant maintain adequate tissue oxygenation>>hypoxia and acidois can cause bradcardia
- This is very immenent for cardiac arresst

CONT'D

- Neonates have limited cardiac reserve; they increase their CO primarily by increasing their HR, rather than SV
- They develop bradycardia as the frst response to hypoxia; older children will initially develop tachycardia

Peripheral perfusion

- SVR may be estimated by examining capillary refill time, skin temperature and diastolic BP
- In the healthy child, the skin is warm, dry and pink from head to toe, unless the ambient temperature is low
- Capillary refll time is used to estimate perfusion; if prolonged, it is an early sign of developing shock

Pulses volume

- SV may be evaluated by palpation of the pulse amplitude
- As SV decreases, pulse amplitude also decreases
- The amplitude of pulses refects the diference between systolic and diastolic blood pressure
- In shock, the pulse wave diminishes in amplitude; it then becomes thready and finally impalpable

Blood Pressure

- The onset of hypotension is particularly late in hypovolaemic shock (occurs after about 40% loss of circulating blood volume)
- With any type of shock, hypotension is sign of physiological decompensation, and must be treated vigorously as cardiorespiratory failure and arrest may be imminent

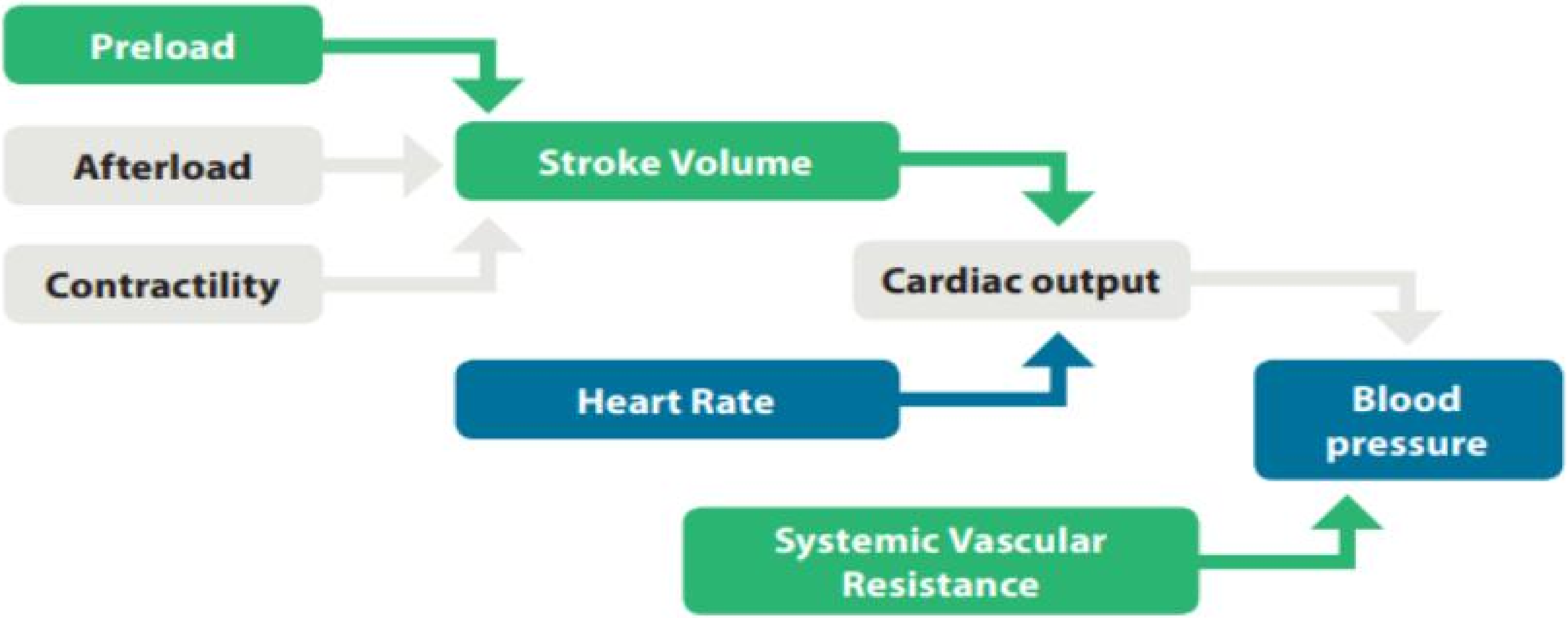
CONT'D

- As it is MAP rather than SBP that determines tissue perfusion, maintaining this MAP above an age specifc threshold is considered at least as important
- A normal MAP can be estimated around [55 + 1.5 *age (years)],
- MAP ≈DBP +1/3 SBP - in cases of severe tachycardia,
- The MAP ≈ 1/2 (DBP+SBP)

PRELOAD

- Its clinical assessment helps to diferentiate cardiogenic shock from other forms of shock
- In children, jugular veins (JV) are visible and edge of the liver can be palpated at maximum 1 cm at inferior margin of rib cage
- In increased preload -as in fuid overload or heart failure-,
- Liver enlarges, JV become dilated and
- Moist sounds (crackles) can be heard in the lungs

Cardiovascular relationships



Relationships between variables affecting cardiac output and variables affecting blood pressure

Variables directly measured

Variables evaluated clinically

Effects on other organs

- Organ perfusion depends on CO and PP. The skin, kidneys, and brain best refect the quality of organ perfusion
  - Tachypnea in hypoperfusion
  - decreased UOP
  - Onset of cerebral hypoperfusion is abrupt
  - Early signs are loss of consciousness, seizures or dilated pupils
  - Then agitation, lethargy, drowsiness or irritability may be observed

Causes of cardiac arrest

- **Reversible causes of cardiac arrest**

4T

Thrombosis
Toxic
Tensionpuemothorax
Tamponade

4H

- Hypoxia
- Hypovolumia
- Hyper/ hypokalaemia (metabolic)
- Hypothermia

Disability - Neurological problems (D)

- Brain and the heart are two organs that are preferentially preserved by the compensatory mechanisms of cardio respiratory failure
- Brain function evaluation to determine child's physiological status
- Following appropriate management of ABC, an evaluation of the child's neurological status should be done (ideally before depresant medication)

CONT'D

- Some neurological conditions like meningitis, status epilepticus or decompensated IC hypertension may have respiratory, circulatory or other severe consequences
- These conditions have to be identifed during the primary assessment since they need to be treated immediately
- A rapid assessment of the child's consciousness can be made using the **AVPU score** (next slide):

CONT'D

- The painful stimulus is delivered by applying sternal or supra-orbital pressure
- A child who is only responsive to painful stimulus has a signifcant degree of neurological derangement equivalent to a value of 8 (which is the cut-of to defne 'Coma') on the Glasgow Coma Scale
- This needs intubation and ventilation

Measuring LOC

D

S C G

- M = 6 obeys commands
- M = 5 localises pain
- M = 4 withdraws to pain
- M = 3 flexes to pain
- M = 2 extends to pain
- M = 1 no motor response

- V = 5 orientated
- V = 4 confused
- V = 3 inappropriate words
- V = 2 inarticulate sounds
- V = 1 no verbal response

- E = 4 eyes spontaneously open
- E = 3 eye opening to speech
- E = 2 eye opening to pain
- E = 1 no eye opening

U P V A

- A – **A**lert
- V - **R**esponds to voice
- P - **R**esponds to pain
- U -**U**nresponsive

GCS ≤ 8
Intubate

Modified GSC for under 5 year children

- Some advocate the use of merely the GCS Motor score as a valuable, more easy to remember alternative
- This score contains nearly all information content of the total GCS
- Deliver pain stimulus by pressing hard on the supra-orbital notch with your thumb, Except for M4
- For M4 press hard on the fat nail surface with the barrel of a pencil
- A motor score of 4 or less indicates ‘coma’

Motor	<5years	>5
M6	Normal spontaneous movements	Obeys commands
M5	Localises to supraorbital pain (> 9 months of age) or withdraws to touch	Localises to supraorbital pain (> 9 months of age) or withdraws to touch
M4	Withdraws from nailbed pain	Withdraws from nailbed pain
M3	Flexion to supraorbital pain (decorticate)	Flexion to supraorbital pain (decorticate)
M2	Extension to supraorbital pain (decerebrate	Extension to supraorbital pain (decerebrate
M1	No response to supraorbital pain (faccid)	No response to supraorbital pain (faccid)

Environment - Exposure

- Proper exposure a child, with respect for their dignity
- Look for presence of fever, signs of trauma, skin rashes etc.
- It is important to avoid hypothermia at any point
- Pain and discomfort should be recognized and treated as soon as possible after the initial ABCDE stabilization
- Look for clues in the environment to understand the problem/diseas

CONT'D

- Use **AMPLE** acronym to explore the circumstances of illness**:**
- **A**llergy?
- **M**edication?
- **P**ast History?
- **L**ast Meal?
- **E**nvironment/exposure?

CONT'D

- Signs of cardiorespiratory failure include altered LOC,hypotonia, tachycardia, decreased central pulses, and absent distal pulses
- Bradycardia, hypotension, bradypnoea, gasping and apnoea are the terminal events that precede imminent cardiorespiratory arrest
- Immediate recognition of circulatory arrest is important, as it demands the initiation of cardiopulmonary resuscitation [CPR]

Basic Life Support

Introduction

- We have pointed out the importance of early recognition and intervention in children who exhibit signs of respiratory and/or circulatory compromise
- Optimal management of respiratory and/or circulatory failure prevents cardiorespiratory arrest
- However, there will always be some children in whom respiratory and/or circulatory arrest cannot be prevented

CONT’D

- For these children, early Basic Life Support [BLS], rapid activation of the Emergency Medical Service [EMS] teams, and prompt, effective advanced life support are crucial in improving mortality and morbidity (Paediatric Chain of Survival)
- BLS is indicated in all children who are unresponsive and not breathing normally

CONT'D

- It must be initiated ASAP, preferably by those witnessing the event
- Its main objective is to achieve sufcient oxygenation to 'protect' the brain and other vital organs
- The sequence of actions in BLS is known as CPR
- BLS is more efective when the rescuer is profcient in its delivery, but even suboptimal CPR gives a better result than no CPR at all

Paeditric BLS sequence

- For the majority of children who sufer cardiorespiratory arrest, the recommended sequence of events is based on two main facts:
  - The majority of paediatric arrests are hypoxic in origin (and/or low FRC), hence priority is to open AW and immediately give oxygen (provided by rescue breaths)
  - Most common cardiac arrhythmia is bradycardia leading to asystole, effective BLS is more important than rapid access to defbrillator

CONT'D

- If witnessed sudden cardiac arrest from VT and/or PVT is the case (occasional), optimal outcome depends on early defbrillation
- It is then preferable for alone rescuer to activate the EMS before initiating BLS and to use an AED, if available

BLS sequence

- Unresponsive>>shout for help>>saftey>>open AW>>not breathing normally>>5rescue breaths>>No signs of life?>>15:2 chest compression to rescue breaths ratio>>call national emergency numbeer after 1 minute of CPR

Paediatric BLS sequence

Safety (S)

- On approaching the child, and before touching him, the rescuer should rapidly look for any clues that may have caused the injury
- This may afect the way the child is managed

Stimulate (S)

- Establish responsiveness of the unconscious child by both verbal and tactile stimulation, as s/he may not be in a critical condition
- Make the head in neutral position, then stimulate on the forehead, hair or arms, no shaking the child
- If there is response by moving, crying or talking, his clinical status and assess any danger signs and, shout if necessary
- If no response, continue with the next steps of BLS

'Stimulate' as part of the BLS sequence

Shout for help (S)

- For only one rescuer, he must shout out for “help” and commence BLS immediately
- He should only leave the child to summon help (or delay BLS to use a mobile telephone) after 1 minute of CPR
- For multiple rescuers, another person should be asked to summon the EMS
  - National EMTnumber 911/907 /939 or in-hospital cardiac arrest team

CONT’D

- A single rescuer will also summon help frst (before CPR) in the case of a sudden witnessed collapse
- If the event is within the healthcare environment, appropriate clinical emergency equipment should also be taken to the scene of resuscitation

Airway (A)

- In unconscious child, the tongue likely obstructs the AW
- Therefore, the rescuer must frst open the airway
- This can be done by performing a Head tilt-chin lift
- If C-spine fructure is suspected the jaw-thrust is preferred
- This latter manoeuvre is however only possible if more than one rescuer is available

CONT'D

- It is important for the rescuer to also have a quick look into the mouth to ensure there is no obvious FB
- If any any iff you are confident to remve it use single finger sweep, if not never try it blidly
- Once the airway (A) has been opened and assessed, the rescuer should move on to the next step

Breathing (B)

- **Assess Breathing**
- The best way to assess breathing is to ‘look, listen and feel‘
- Maximum time allowed for this should be 10 seconds
- If the child is breathing spontaneously and efectively, the airway is kept open whilst more assistance is summoned

CONT’D

- If the child is not breathing efectively or only gasping,
- The rescuer must deliver rescue breaths
- Gasping or agonal breathing is infrequent or irregular breaths that must not be confused with normal breathing

CONT'D

- **Rescue breaths**
- Give five initial rescue breaths while maintaining the airway open
- The aim is to deliver oxygen to the child's lungs
- Each breath should be delivered slowly over about 1 second
- The rescuer must take a breath between each rescue breath to optimise the O_2 and minimise the CO_2 they deliver to the child
- How would you see effectiveness of this rescu breaths?

CONT'D

- **Infants:** Mouth to mouth-and-nose technique
- **Child:** Mouth-to-mouth technique
- Once the initial rescue breaths have been delivered, the rescuer should move onto circulation (C)

CONT'D

CIRCULATION (C)

- **Assess for ‘signs of life’**
- Observe the child for ‘signs of life-movement, coughing or normal breathing only for10 seconds
- Gasps or infrequent, irregular breaths are abnormal
- If there is no ‘signs of life’ start chest compression
- If there are signs of life, the rescuer should reassess the child’s breathing and continue the rescu breaths at RR for age

CONT'D

- The childs breathing and circulation are frequently reassed and rescucitation should contineu until EMS arrives
- Only if efective spontaneous breathing is clearly established and there is no suspicion of cervical spine trauma, the child can be placed in a safe, 'recovery position

CONT’D

- **Chest Compressions**
- To delivere efectively, child must be placed supine on a hard, fat surface, maintaining head in a position that keeps the airway open
- At a rate of 100-120 times per minute
- The depth at least 1/3 anteroposterior diameter of the chest
- Keep compression: Ventilation ratio of 15:2 for all age children

Ratio of Ventilation to Compression

1. Adult: 30 compressions to 2 ventilations
2. Child: 15 compressions to 2 ventilations
3. Newborns: 3 compression to a ventilation

CONT’D

- Allow for chest recoil after each compressions
- Minimize interruption
- Deliver chest compressions over the lower half of the sternum

Reassess (R)

- After one minute or four cycles of CPR, briefy stop CPR to reassess the child by quickly looking for 'signs of life'
- If the EMS team has already been alerted, continue BLS, only stopping if indicated

CONT'D

Chest compression in infant: two-finger technique

Chest compression in infant: two-thumb encircling technique

When to stop BLS ?

- BLS should only be stopped when:
  - ➢The child exhibits signs of life
  - ➢Other rescuers (EMS) take over resuscitation
  - ➢The safety of the rescuer can no longer be guaranteed
  - ➢The rescuer becomes too exhausted to continue

Sudden witnessed collapse

- This may be a sudden arrest without any prior signs and sypmtoms as witnesed by others
- The most likely cause is a malignant arrhythmia
- Early defbrillation takes a frst priority, a single rescuer would call first (instead of after 1 min) and
- If available apply an AED as soon as possible
- Rescuers familiar with adult algorithm could prefer to use this

CONT'D

- Use of an AED in children
- The AED analyses the victim's ECG and determines the need for, and then if appropriate, delivers an asynchronous electrical shock (with a predefned amount of Joules)
- All new devices are biphasic(monophasics are also in use)
- For a child over 25 kg (>8 yrs) standard 'adult' AED can be used

CONT'D

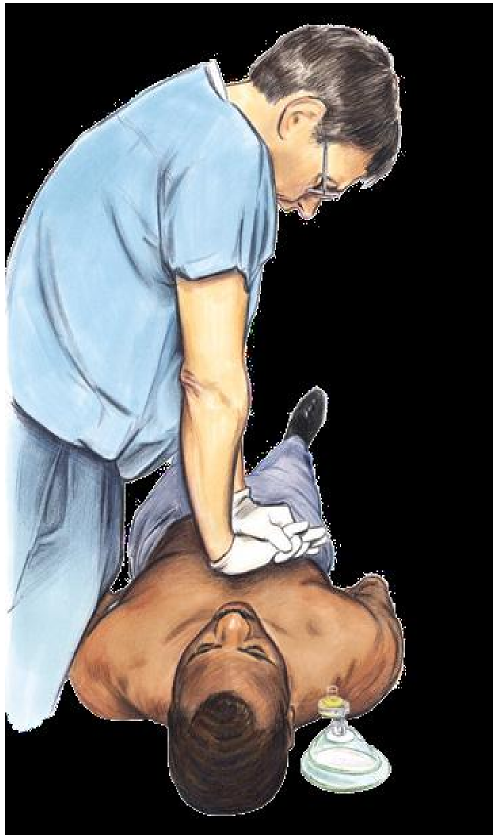
- Ideally, use AED with attenuating device for children <8 years
- This delivers a lower energy 50-75J (tthan 150-200J if biphasic)
- Children aged <1 year have a much lower incidence of shockable rhythms and
- The focus of resuscitation in infants should be good quality CPR

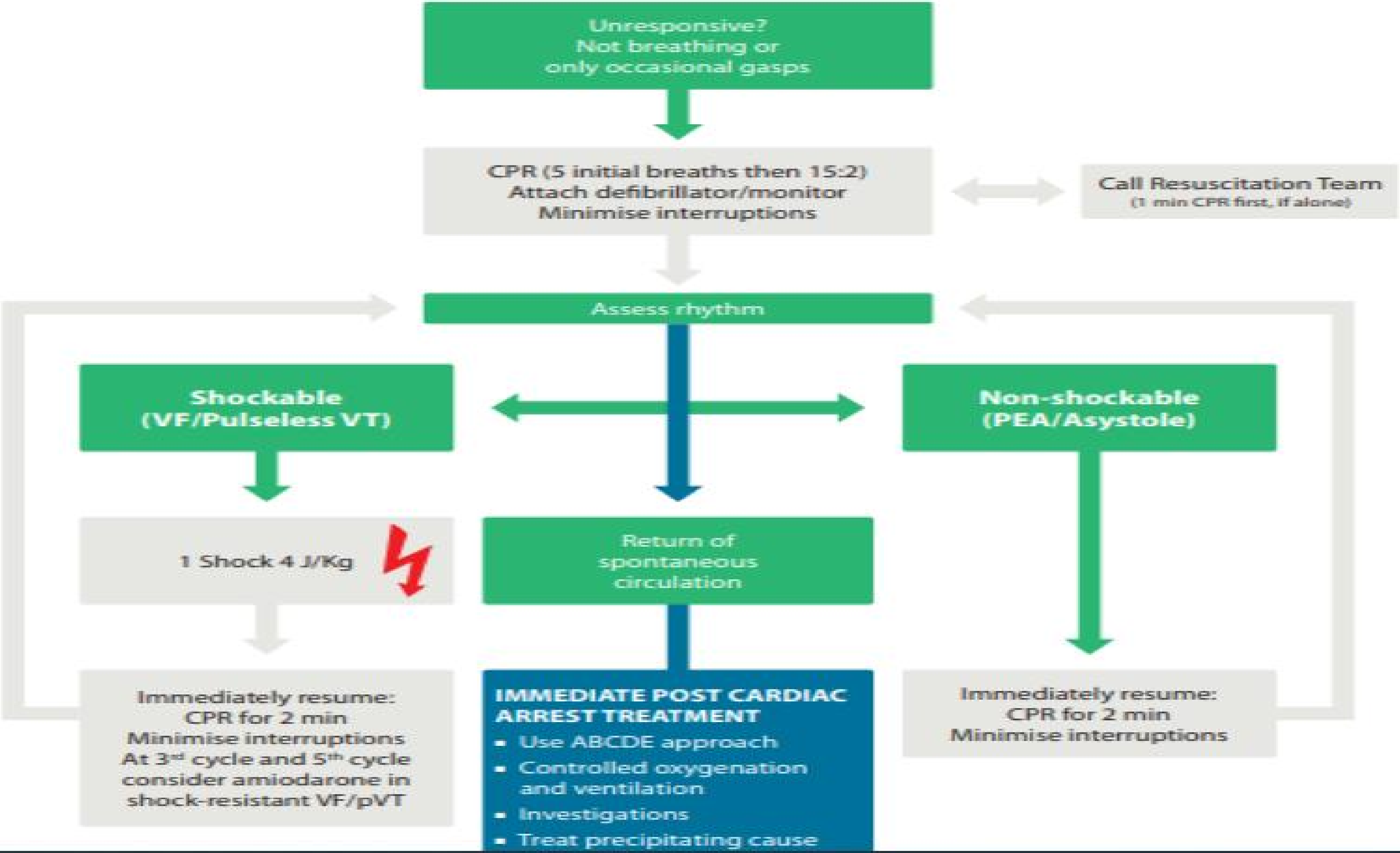
CONT'D

- AED use should not delay the delivery of good quality CPR
- In cases where there are two or more rescuers, CPR should not be interrupted while the adhesive electrode pads are being placed and the child should receive at least one minute of high quality BLS before the AED is switch on

Qualified CPR

- Push hard (4-5cm) & push fast (100 to 120/m)
- Full chest recoil,
- Minimize interruptions
- Avoid excessive ventilations
- Rotate compressors every 2 min
- 15:2 ratio (if no advanced AW)





Unresponsive?
Not breathing or
only occasional gasps
CPR (5 initial breaths then 15:2)
Attach defibrillator/monitor
Minimise interruptions
Call Resuscitation Team
(1 min CPR first, if alone)
Assess rhythm
Shockable
(VF/Pulseless VT)
Non-shockable
(PEA/Asystole)
1 Shock 4 J/Kg
Return of
spontaneous
circulation
Immediately resume:
CPR for 2 min
Minimise interruptions
At 3rd cycle and 5th cycle
consider amiodarone in
shock-resistant VF/pVT
IMMEDIATE POST CARDIAC
ARREST TREATMENT
Use ABCDE approach
Controlled oxygenation
and ventilation
Investigations
Treat precipitating cause
Immediately resume:
CPR for 2 min
Minimise interruptions

BLS algorithm with AED

1. Safety for the rescuer, the victim, and any bystander
2. Commence appropriate BLS
3. Place AED pads as one to the right of sternum, below the clavicle and the other in the mid-axillary line on the left side of the chest

CONT'D

4.Switch on the AED. End CPR cycle with compressions

5. Follow the spoken/visual directions

6. Ensure that nobody touches the victim while the AED is analysing the rhythm

7. If a shock is indicated: ensure that nobody touches the victim and then push the shock button as indicated.Then, immediately resume appropriate BLS

CONT'D

8. If no shock is indicated: immediately resume the BLS

9. Continue as directed by the voice/visual prompts.

10. Continue CPR until one or more of the following occurs:

   – Further help arrives and takes over

   – The victim starts to breathe normally

   – You become exhausted

PEADIATRICS ADVANCED CARDIAC LIFE SUPPORT (PALS)

Goal of CPR

CPR- is given for patients who have stopped breathing and heart beat.

- The goal of resuscitation is
  - Restore effective oxygenation & ventilation
  - Restore circulation
  - Return of intact neurological functions

CONT'D

Chain of survival

- For effective result of resuscitation there should be:
  1. Early access to the patient or victim
  2. Early CPR initiation
  3. Early defibrillation
  4. Early & effective post resuscitation care

The Paediatric Chain of Survival identifies crucial factors to achieve optimal outcome.

Consequences of cardiac arrest

- Rapid depletion of oxygen in vital organs
- After 4-6 min of cardiac arrest brain damage can occur
- Resuscitation beyond 10min of arrest is associated with poor outcome
- Early CPR within 4 min and rapid ALS with defibrillation within 8 min are essential in improving survival and neurological recovery

CONT’D

- Artificial distinction between BLS and PALS
- The process of resuscitation started with BLS (chest compressions and ventilations) must be continued until ROSC
- Then followed by adequate post-resuscitation care (paediatric chain of survival)

CONT'D

- All clinical staff within a healthcare facility should be able to immediately recognise cardic arrest, commence appropriate resuscitation and summon an ALS team as appropriate

Manual defbrillators

- Manual defbrillators are capable of delivering the full energy requirements from neonates upwards
- They have several advantages over AEDs and
- It must be readily available in all HF, even if AEDs are ready

Advantages over the AED

- Arrhythmia diagnosis and if necessary, rapid shock delivery with less hands-of time than waiting on rhythm analysis by an AED
- Additional treatment possibilities, such as synchronised cardioversion and external pacing
- Ability to alter energy levels
- A continuous ECG monitoring

Defibrillation

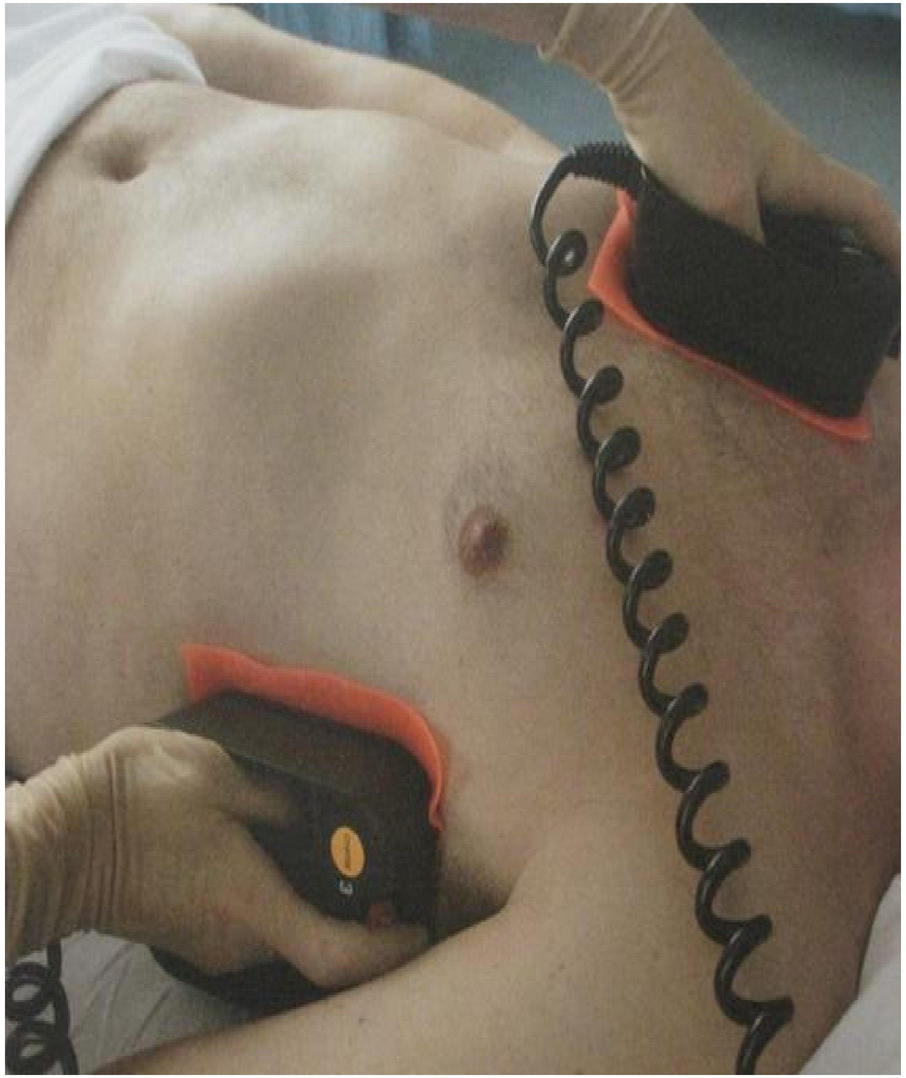
- Is delivery of electrical current across the myocardium with the intention of causing a global depolarisation of the myocardium and restoration of organised spontaneous electrical activity
- The energy dosage causes minimal myocardial injury
- The electrical current depends on the selected energy (Joules) and resistance to fow of current (thoracic impedance)
- If impedance is high, the energy must be increased

CONT’D

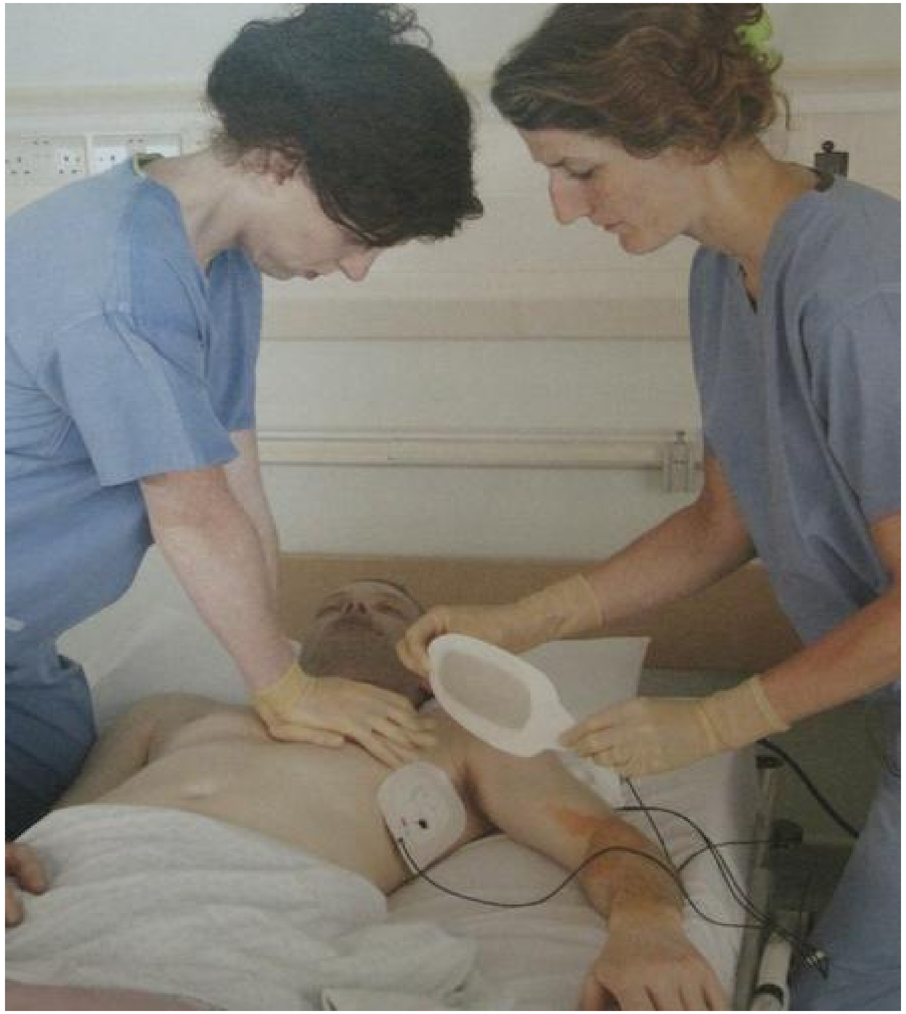
- Factors determining thoracic impedance are:
  - Paddle/pad size
  - Conduction interface between paddle and skin
  - Number and time interval of previous shocks
  - Positioning of the paddles on the chest wall
  - Paddle electrode pressure on the chest
  - Chest wall thickness and obesity

ENERGY LEVELS

- Monophasic Vs biphasic(more effective) defbrillators
- Ideal energy dose for effective defibrillation in children(unknown
- But, evidences advise to select an energy dose of 2-4J/kg for all shocks when using a manual defbrillator, monophasic or biphasic
- Max.dose for first shock is 200J (biphasic) & 360 J (monophasic)



Standard electrode positions for defibrillation



Standard electrode positions for defibrillationIf self-adhesive pads are used, maintain chest compressions

Safety during defbrillationa

- Remove any free-fowing oxygen devices
- Dry surfaces
- Contact – Ensure every body is clear of patient contact
- Ensure the pads/paddles are not in contact with metals
- The pads/paddles should be placed at least 12cm far from cardiac pacemakers or implantable cardioverter-defbrillators (ICD)

CONT'D

- Ensure that the operators are familiar with its use
- Ensure clear safety instructions to the team/bystanders
- Any defbrillator should be tested regularly, as per local policy

Advanced life support in paediatric cardiorespiratory arrest

Initiation of BLS

- **Safety (S):** regardless of the setting ensure PPE ASAP
- **Stimulate (S)**: responsiveness of an unconscious child should be ensured either by verbal or tactile gentle stimulation
- **Shout for help (S)**: a single rescuer shouldn't leave the collapsed child but immediately initiate resuscitation as appropriate and ensure that further help is summoned

CONT'D

- **A:** the airway should be opened, use **LLF**
- **B:** if the child is apneic or gasping - give initial rescue breath by most appropriate means
- **C:** if no signs of circulation, chest compressions must be initiated
- Minimise interruptions by changing roles in a timely manner to avoid periods of no resuscitation
- Attention should be given at all times to the quality of CPR

CONT'D

- In infants, the recommended sites for assessing the central circulation are the brachial or femoral artery
- In children, it is the carotid or femoral artery
- Absence of a palpable central pulse or a pulse that is very slow (< 60 beats/min) with signs of decompensated shock indicates circulatory arrest. In doubt, start CPR

Rhythm recognition

- While CPR is performed, the next step is identifing cardiac rhythm and therefore an ECG monitor or
- A defbrillator must be attached to the patient
- This to decide whether cardiac rhythm is shockable or not, in order to determine the next steps in the CPR management
- If AED is in used, the machine will guide the rescuers through the appropriate sequence of actions

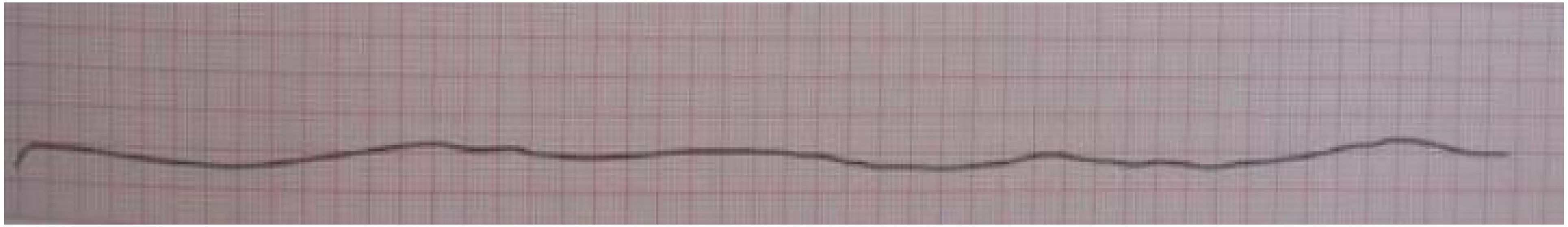
CONT’D

- The commonest initial cardiorespiratory arrest rhythms in children are non-shockable rhythms: PEA and asystole
- Mng’t priorityt for shockable rhythms is early defbrillation
- PEA and pVT are potentially perfusing rhythms
- The need for CPR in these rhythms is identifed by the absence of signs of life (and optionally a central pulse)

Normal sinus rhythm

Asystole –Non-shockable

- Asystole can be simulated by an artifact (detached electrodes)
- A prompt check of equipment and selected ECG lead/paddle, is essential to eliminate these possible artifacts



PEA

- In a patient with no pulse and no sign of life

- Pulseless electrical activity –Non-shockable

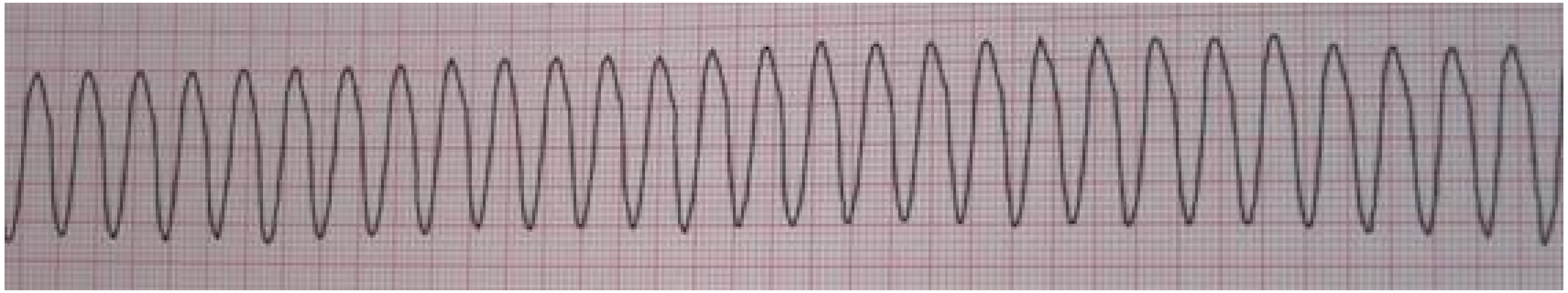
VF

ventricular fibrillation

VF

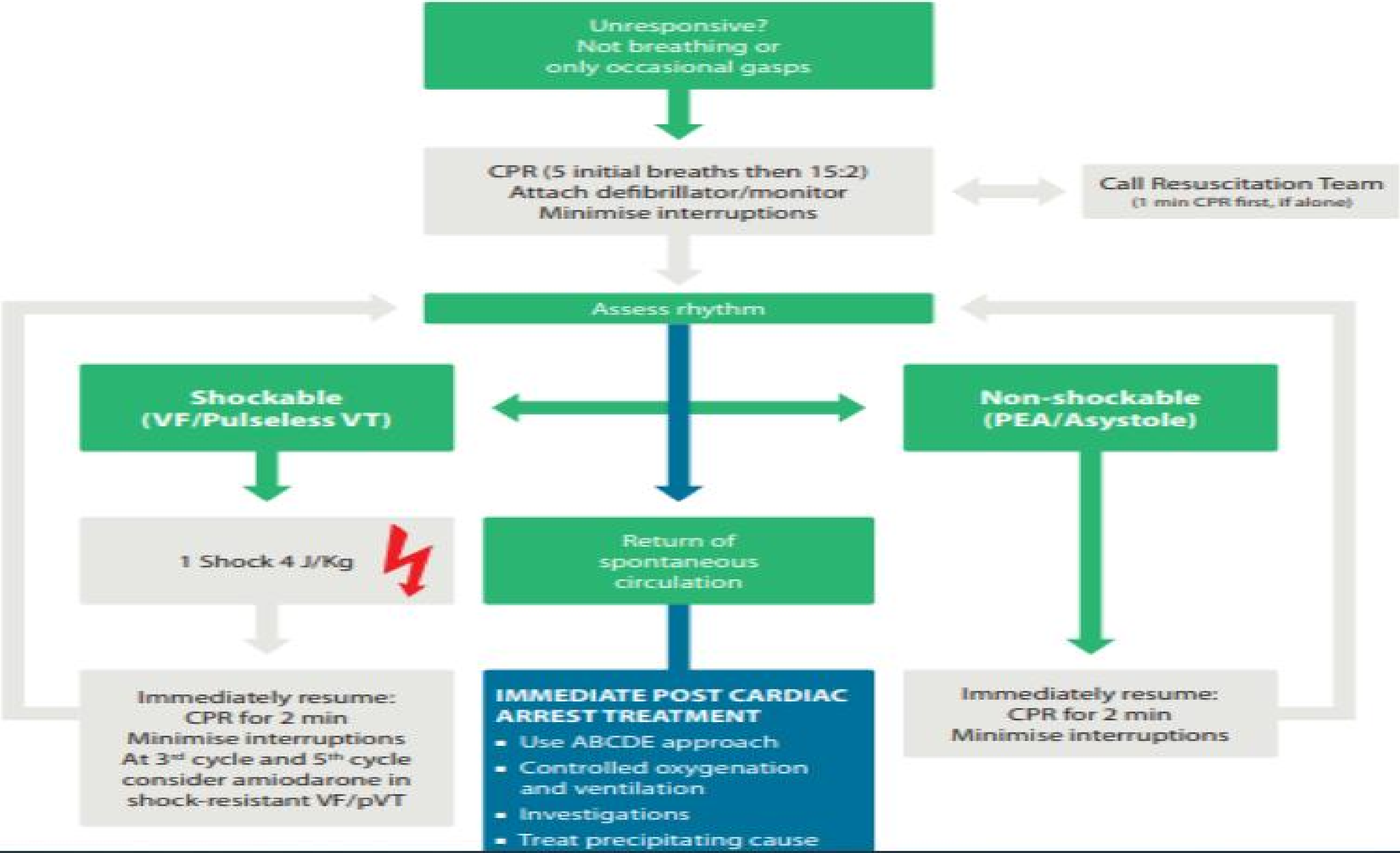
- When there is a doubt whether a rhythm is a fine VF or an asystole, rescuers should continue with CPR
- It is unlikely that fine VF could be successfully shocked into a perfusing rhythm
- High quality CPR may increase amplitude and frequency of the VF, which improves the chances of successful defbrillation

VT



Ventricular tachycardia

Ventricular Tachycardia is ”shockable” (remember to check for a pulse, it may be compatible with a CO)


Unresponsive?
Not breathing or
only occasional gasps
CPR (5 initial breaths then 15:2)
Attach defibrillator/monitor
Minimise interruptions
Call Resuscitation Team
(1 min CPR first, if alone)
Assess rhythm
Shockable
(VF/Pulseless VT)
Non-shockable
(PEA/Asystole)
1 Shock 4 J/Kg
Return of
spontaneous
circulation
Immediately resume:
CPR for 2 min
Minimise interruptions
At 3rd cycle and 5th cycle
consider amiodarone in
shock-resistant VF/pVT
IMMEDIATE POST CARDIAC
ARREST TREATMENT
Use ABCDE approach
Controlled oxygenation
and ventilation
Investigations
Treat precipitating cause
Immediately resume:
CPR for 2 min
Minimise interruptions

DURING CPR

- Ensure high-quality CPR: rate, depth, recoil
- Plan actions before interrupting CPR
- Give oxygen
- Vascular access (intravenous, intraosseous)
- Give adrenaline every 3-5 min
- Consider advanced airway and capnography
- Continuous chest compressions when dvanced airway in place
- Correct reversible causes

REVERSIBLE CAUSES

- Hypoxia
- Hypovolaemia
- Hyper/hypokalaemia, metabolic
- Hypothermia
- Thrombosis (coronary or pulmonary)
- Tension pneumothorax
- Tamponade (cardiac)
- Toxic/therapeutic disturbances

ACTIONS FOR NON-SHOCKABLE RHYTHM

- Continue high-quality CPR at 15:2 ratio with BMV and O_2
- Minimise interruptions and avoid hyperventilation
- Obtain IV access (IV or IO), IO route has become frst choice in paediatric cardiac arrest, if no line is in place
- Administer adrenaline at 0.1 ml/kg of a 1:10000 solution with a max 1 mg or 10 ml then 2-10 ml fush of N/S

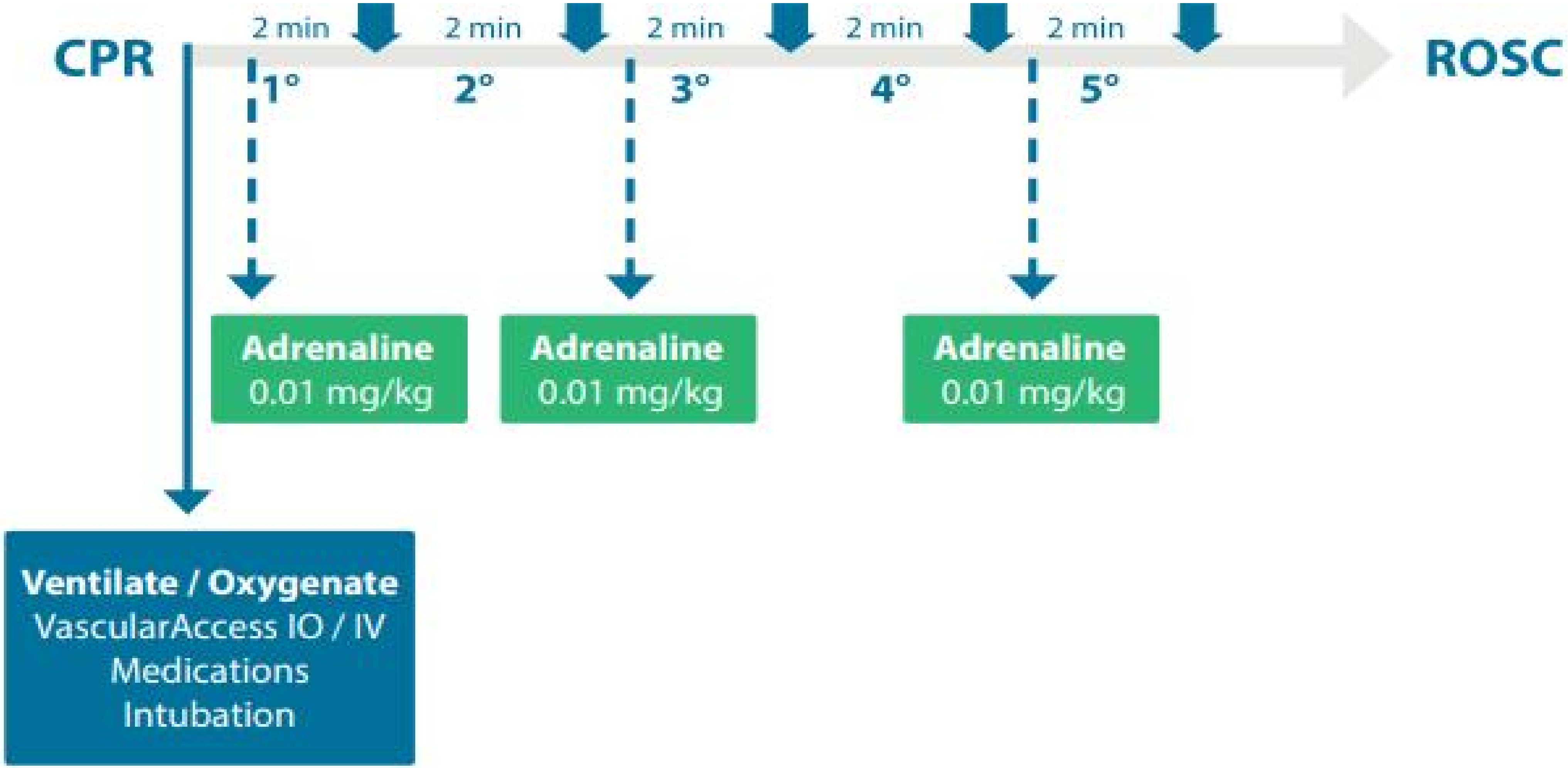
CONT'D

- After 2 min of CPR: reassess rhythm at 2 vent or in continuous compressions, a very brief pause in chest compressions
- If rhythm is asystole or no change in the ECG, resume CPR immediately
- If an organised electrical activity (which is a potentially perfusing rhythm) is seen on the monitor (check for signs of life and a central pulse)

CONT'D

  - If signs of life are present, start post-resuscitation care
  - If not, continue CPR (minimising interruptions)
- IV/IO adrenaline may be repeated every 3-5 minutes

Non-shokable algorism



ACTIONS FOR A SHOCKABLE RHYTHM

- Defibrillation should be done ASAP if there is a defbrillator
- Ventilation,oxygenation, chest compression, and vascular access should be quickly instituted but without delaying defbrillation
- Minimise interruptions in chest compressions by planning actions
- Minimise rescuer fatigue by changing role every 2min of CPR
- Do not interrupt CPR more than 5 seconds for any action

Sequence of actions when using self-adhesive pads (first choice)

1. Confrm cardiac arrest and then start CPR
2. Place appropriate self-adhesive pads on the child's chest
3. Switch the defbrillator on, ensure it is in the non-synchronised mode; select 'pads' as monitor lead and confirm shockable rhythm during a brief pause in chest compressions, for instance during the 2 given ventilations

CONT'D

4. Select a 4 J/kg energy and charge the defbrillator without interrupting compressions

5. Say: "STAND CLEAR, SHOCKING" in a loud voice, confirm no one touches a patient O2 sources are D/C

6. Deliver the shock

After the first shock

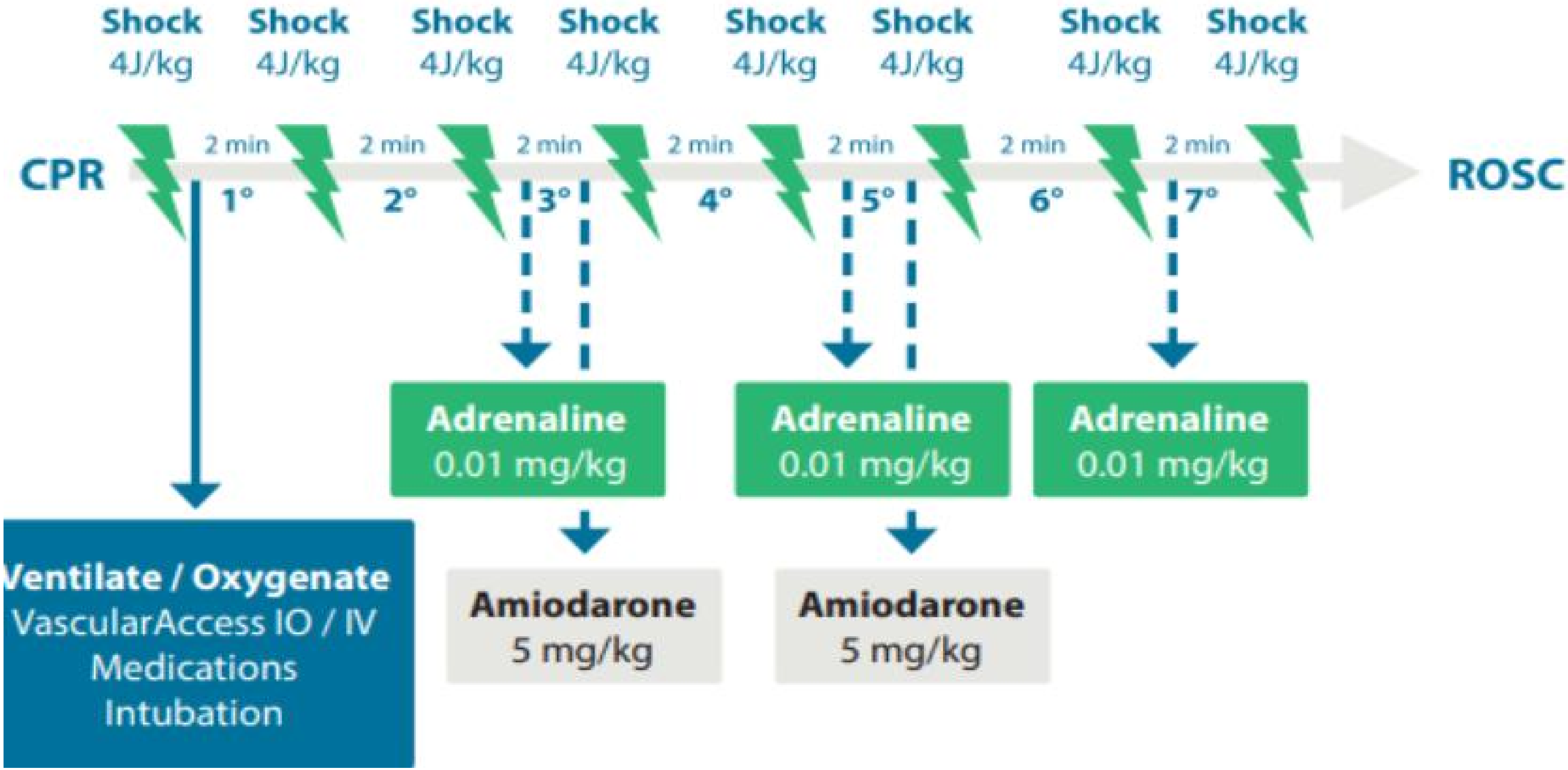
7. Request immediate resumption of CPR

8. Continue CPR for 2 minutes then briefy pause to check the ECG. If the rhythm is still a shockable, repeat previous steps and deliver a second shock. Then continue CPR for 2 minutes after which, if VT/VF persists, a third shock is indicated

9. After the third shock, once CPR has been resumed, administer adrenaline 10 mcg/kg IV/IO and amiodarone 5 mg/kg IV/IO

CONT'D

- Give adrenaline every alternate cycle (every 3-5 minutes) during CPR
- If the child remains in VF/pVT, continue to alternate shocks of 4 J/kg with 2 min of CPR
- Give a second (final) dose of amiodarone 5 mg/kg after the 5th shock if VF/VT persists



Shock 4J/kg
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Shock 4J/kg
CPR
2 min
2 min
2 min
2 min
2 min
2 min
2 min
ROSC
1°
2°
3°
4°
5°
6°
7°
Adrenaline 0.01 mg/kg
Adrenaline 0.01 mg/kg
Adrenaline 0.01 mg/kg
Amiodarone 5 mg/kg
Amiodarone 5 mg/kg
Ventilate / Oxygenate
VascularAccess IO / IV
Medications
Intubation

Note

- If a potentially perfusing rhythm is seen after 2-minute cycle of CPR the following actions should be taken:
  - If there is signs of life, start post-resuscitation care
  - If there are no signs of life, resume CPR and if PEA, switch to the non-shockable algorithm

CONT'D

- If the rhythm changes to asystole after 2-min cycle of CPR:
  - Resume CPR and switch to the non-shockable side
  - When switching from shockable to non-shockable algorithm maintain 3-5 minute intervals between adrenaline administration

Decision making

- Confrmation of cardiorespiratory arrest
- Calling the resuscitation team
- Starting CPR
- Attaching a defbrillator and delivering a shock

Team working

- Team leader
- Team members

Task management

- Diagnosis of the cardiorespiratory arrest rhythm
- Choice of shock energy to be used for defbrillation
- Likely reversible causes of the cardiorespiratory arrest
- How long to continue resuscitation

Performing CPR as a team

- Preparation/Anticipation
- Resuscitation
- Post Resuscitation Care

CONT’D

- Resuscitation does not stop with ROSC
- Handing the patient over to another colleague, or dep’t or to a diferent hospital requires good communication and handover tools can provide a framework for information sharing at this stage
- The team leader will decide when to stop resuscitation efforts

When to stop rescucitation

1. With return of spontaneous circulation
2. No ROSC during or after 20 minutes of resuscitative efforts
   - Possible exceptions include
     - near-drowning,
     - severe hypothermia
     - known reversible cause
     - some overdoses (LA toxicity)
3. DNR orders presented
4. Obvious signs of irreversible death